Description

Illustration





Figure caption

1. Decorative ceramic tiles used as flooring 2. Glazed tiles are an ideal material for roofs and other assemblies for catching and directing water. Images used under license from Shutterstock.com

The material

Tiles are widely used in building as cladding, covering or finish for roofs, walls and flooring where they provide weather protection, resistance to wear, and decoration. Tiles have, for millennia, been made by baking and glazing clay ("ceramic tiles"), but today fiber composites, thermoplastics (UPVC for example) and reconstituted stone are also formed into tiles. Ceramic tiles are among the most durable of building materials. They weather well, resist attack by most acids and alkalis, are impervious to oils, grease and cleaning materials, and are abrasion resistant. Glazing can be used to give vibrant colors and patterns, exploited to ultimate effect in mosaics and wall tiling, as bright today as when they created as much as 2000 years ago.
Tiles are typically manufactured in relatively small sizes allowing for ease of construction in the making of intricate patterns and the application to complex surfaces. Extraordinary examples include royal palaces, civic architecture, residential buildings and others of many countries including Japan, Spain (the Alhambra for example), India, China, Italy and many others.

Compositional summary

Clay is a naturally occurring alumino-silicate (Al2O3-SiO2). The composition given here is the mean of the three alumino-silicates andalusite, sillimanite and kyanite: 63% Al2O3 / 36.7% SiO2 / 0.27% Fe2O3 / 0.027% K2O / 0.016% MgO

General properties

Density	2.05e3	-	2.4e3	kg/m^3
Price	* 0.393	-	0.43	GBP/kg
Date first used	-3000			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✗
Enclosure	✓
Interiors	✓
Services	✗

Mechanical properties

Young's modulus	* 4	-	8	GPa
Shear modulus	* 1.67	-	3.33	GPa
Bending modulus	* 4	-	8	GPa
Poisson's ratio	* 0.196	-	0.204	
Yield strength (elastic limit)	3	-	6	MPa
Tensile strength	* 3	-	6	MPa
Compressive strength	* 20	-	50	MPa
Bending strength	* 3	-	6	MPa
Elongation	* 0.04	-	0.15	% strain
Hardness - Vickers	* 8	-	30	HV
Fracture toughness	* 1	-	2	MPa.m^0.5
Mechanical loss coefficient (tan delta)	* 0.001	-	0.003	

Thermal and combustion properties

Thermal conductor or insulator?	Poor insulator			
Thermal resistivity	1.18	-	1.33	m.°C/W
Thermal expansion coefficient	* 5	-	8	µstrain/°C
Specific heat capacity	* 750	-	850	J/kg.°C
Melting point	* 1.2e3	-	1.5e3	°C
Maximum service temperature	* 673	-	1.08e3	°C
Flammability	Non-flammable			
Emissivity	0.5			

Hygro-thermal properties

Water absorption	* 0.05	-	0.3	%
Water vapor permeability	1.02e-13	-	7e-13	kg.m/s.m^2.Pa
Frost resistance	Average			

Electrical properties

Electrical conductor or insulator?	Good insulator			
Electrical resistivity	1e14	-	3e14	µohm.cm

Dielectric constant (relative permittivity)	4	-	10	
Dissipation factor (dielectric loss tangent)	0.001	-	0.01	

Optical properties

Natural color

The matrix material of tiles is dependent on the clays used and the processes used to fire them. Glazing provides the opportunity for coloring the surface and applying texture.

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	Yes
Notes	Al (aluminum) added to the 2018 US critical minerals list

Acoustic properties

Sound absorption	Poor
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Moldability	2	-	4	
Machinability	1	-	2	

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Excellent
Strong acids	Excellent
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Excellent
Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 1.59	-	1.76	kg/kg	
Embodied energy (virgin grade production)	* 11.4	-	12.6	MJ/kg	

Water Usage	* 12	-	13.2	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 17.4	-	22.8	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.14	-	1.49	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.159	-	0.176	kg/kg
Embodied energy, recycling	* 1.6	-	1.76	MJ/kg
Recycle fraction in current supply	0			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

The proper use of tiles in buildings is primarily dependent on the interaction between the tile and the adhesive materials used to attach the tile to a planar surface and the grout used to seal the joints between them. Cementitious grouts are most common when used with tiles meant to produce a watertight surface for drainage from a shower stall or washroom. Silicone is also used as a sealant in these kinds of applications. Mechanical restraint is rare but used when tiles form the "rainscreen" assembly of an exterior wall (see Terracotta).

Technical notes

Tiling is covered by the following standards: BSCP 5385, BSCP 202, BSCP 5534 and BS 8203.

Typical uses

interior floors; walls in bathrooms and other 'wet' rooms; pools; laboratories; exterior building enclosures and roofs, paving, and other weather-resistant surface applications.

Tradenames

Thousands of companies (using thousands of tradenames) produce tiles for use in architecture.

Related standards

BS 402

Further reading

Doran, D.K. (ed.) Construction materials reference book, Butterworth Heinemann, Oxford (1992)

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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